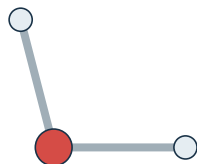


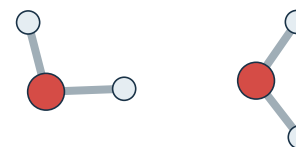
Water clusters | Python vs LAMMPS

2 isolated, neutral, unoptimized geometries · single-point energies and forces

Monomer



Distorted Dimer



TOTAL ENERGIES / kcal/mol; ΔE = Python – LAMMPS

Structure	Atoms	Python total	LAMMPS total	ΔE
monomer	3	-248.344209206	-248.344209206	+1.421e-13
distorted dimer	6	-498.807203052	-498.807203052	+3.411e-13

FORCE DIFFERENCES / kcal/mol/Å; Cartesian components

Structure	Fixed-charge max $ \Delta F $	Fixed-charge RMS ΔF
monomer	4.086e-14	2.237e-14
distorted dimer	4.241e-13	1.537e-13

Fixed-charge forces: QEq at each structure; charges held constant only during differentiation (default).

All differences subtract LAMMPS forces. RMS averages all 3N Cartesian components.

Reference: LAMMPS (22 Jul 2025 - Update 4) via Imp_mpi. Parameters: ffield.reax.HO.2015; Achtyl et al. (2015).

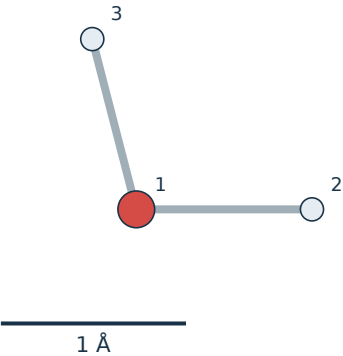
Parameter SHA-256: 526dea2d55d748caed94734139dda66064e194f74f46eee9b0f511eead38b7d1

XREAC / Water-cluster reference validation

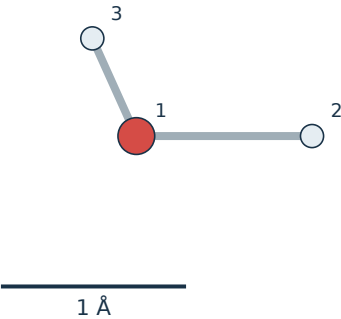
Monomer

3 atoms · O in red, H in pale gray · labels follow XYZ atom order · lines: bond order > 0.3

XY projection



Tilted projection (55° about x)

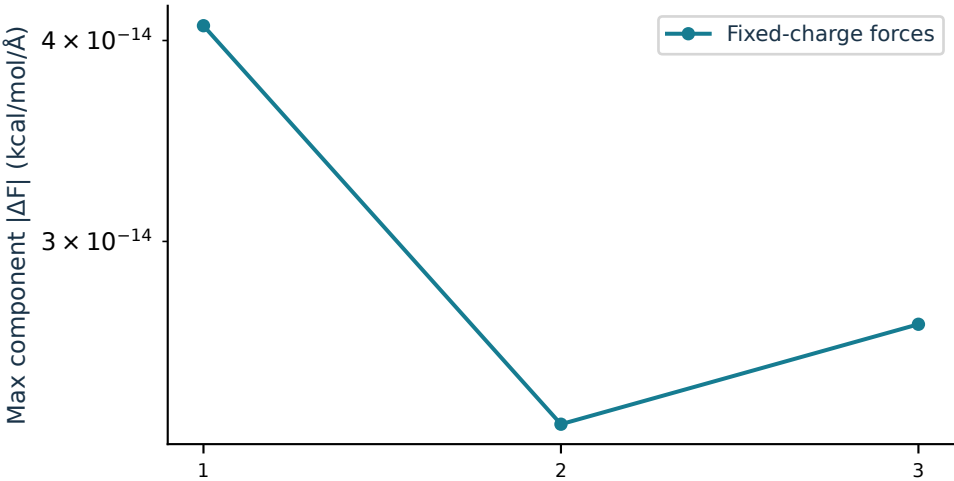


Quantity	Value
Python E (kcal/mol)	-248.344209206337
LAMMPS E (kcal/mol)	-248.344209206337
ΔE (kcal/mol)	+1.421085e-13
Fixed-charge max $ \Delta F $	4.085621e-14
Fixed-charge RMS ΔF	2.236768e-14

ENERGY COMPONENTS / kcal/mol

Component	Python	Δ vs LAMMPS
bond	-256.95789505	+0.00e+00
atom	-9.48962027	+0.00e+00
lone pair	-0.00000000	+0.00e+00
molecule	0.00000000	+0.00e+00
angle	3.08139970	+0.00e+00
penalty	0.00000000	+0.00e+00
angle conjugation	0.00000000	+0.00e+00
hydrogen bond	0.00000000	+0.00e+00
torsion	0.00000000	+0.00e+00
conjugation	0.00000000	+0.00e+00
vdw	48.67028283	+0.00e+00
coulomb	-78.87661327	+9.95e-14
electric field	0.00000000	+0.00e+00
qeq	45.22823685	+2.84e-14

Force differences from LAMMPS



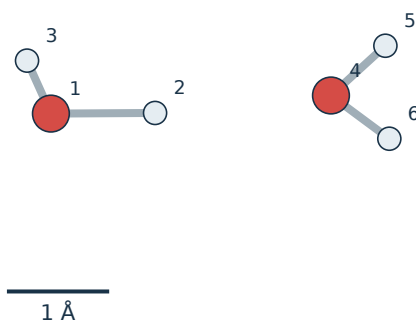
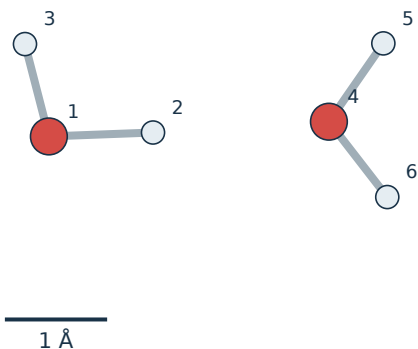
Each point: max over x, y, z for one atom.
Log scale; values below 10^{-16} shown at 10^{-16} .
Force summary above uses kcal/mol/Å throughout.

Distorted Dimer

6 atoms · O in red, H in pale gray · labels follow XYZ atom order · lines: bond order > 0.3

XY projection

Tilted projection (55° about x)

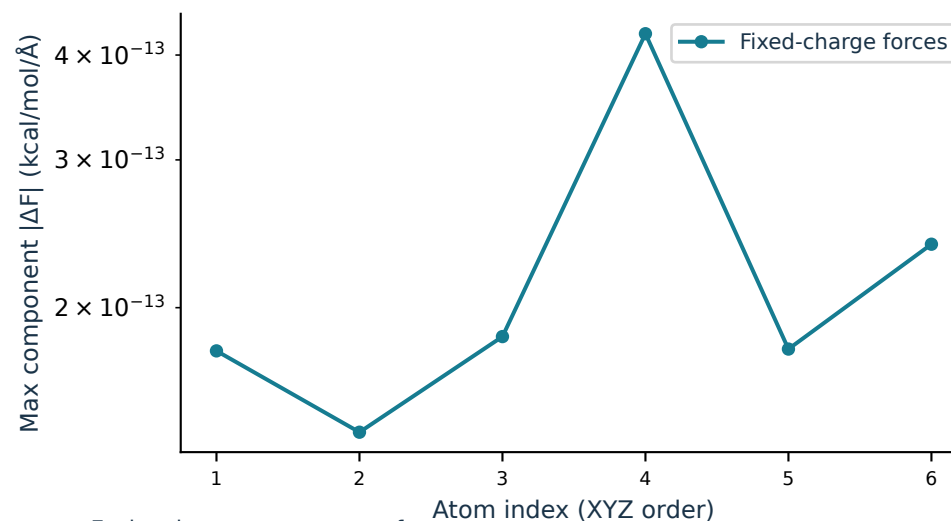


Quantity	Value
Python E (kcal/mol)	-498.807203051698
LAMMPS E (kcal/mol)	-498.807203051698
ΔE (kcal/mol)	+3.410605e-13
Fixed-charge max $ \Delta F $	4.241052e-13
Fixed-charge RMS ΔF	1.537036e-13

ENERGY COMPONENTS / kcal/mol

Component	Python	Δ vs LAMMPS
bond	-507.79364459	+0.00e+00
atom	-17.85610067	+0.00e+00
lone pair	-0.00000001	+0.00e+00
molecule	0.00000000	+0.00e+00
angle	7.00067788	+1.78e-15
penalty	0.00000000	+0.00e+00
angle conjugation	0.00000000	+0.00e+00
hydrogen bond	-1.45319620	-2.22e-16
torsion	0.00199824	+1.65e-17
conjugation	0.00000000	+0.00e+00
vdw	92.27169959	+0.00e+00
coulomb	-181.08981660	+1.31e-12
electric field	0.00000000	+0.00e+00
qeq	110.11117930	-1.04e-12

Force differences from LAMMPS



Each point: max over x, y, z for one atom.
Log scale; values below 10⁻¹⁶ shown at 10⁻¹⁶.

Force summary above uses kcal/mol/Å throughout.